Neural Signal Decoding Using EEG and EOG Channel Signals

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***Abstract*—Motor imagery classification uses electroencephalograph (EEG) signals to develop brain-computer interface (BCI) systems, but is constrained by the variability and time complexities associated with neural data. In our study, we analyze the effects of different feature representations and models on the accuracy of classification algorithms. Various traditional machine learning (ML) algorithms, such as logistic regression (LR), support vector machines (SVM), and random forests (RF), were compared with deep learning (DL) algorithms, such as 1D and 2D convolutional neural networks (CNNs). Feature-based ML approaches considered flattened EEG signals, statistical features, and bandpower representations, while representation learning approaches used 1D CNN for direct modeling of time-dependent structures in neural data and 2D CNNs with Gramian Angular Fields (GAF) transformations for modeling of images. As shown by experiments, proposed deep learning models outperform all traditional approaches, with the enhanced 1D CNN demonstrating the best performance. In turn, 2D CNNs with GAF representations also produce high-quality results, making GAF transformations useful tools to consider in the classification of motor imagery from EEG signals, whereas simplified statistical features are shown to lead to decreased performance, which might be attributed to the loss of data.**

***Index Terms*—Motor imagery, EEG, Brain-Computer Interface, Convolutional Neural Networks, Machine Learning**

I. Introduction

BCI systems aim to convert brain activity into machine-interpretable signals that enable communication between the brain and external devices via mental commands, without requiring any body movements. As a prominent technology for BCI systems, EEG is chosen for its non-invasive nature, high temporal resolution, and affordability. One of the most critical applications of EEG-based BCIs is motor imagery classification, which involves building a model to identify intended actions (such as moving the left or right hand) from recorded brain signals alone. Motor imagery decoding is an essential component of BCI-based assistive technologies, particularly for those with motor disabilities.

Despite its advantages, EEG still faces challenges due to poor signal-to-noise ratio, non-stationarity, and high inter-individual variability. The pipeline of an EEG-based BCI system consists of signal preprocessing, feature extraction, and classification. Each step is vital in contributing to successful classification. Conventional strategies rely heavily on manually constructed features, including frequency-band power, time-domain measures, and spatial filters [1]. Among these methods, CSP (common spatial patterns) feature extraction is among the most frequently employed for EEG-based motor imagery classification, where CSP optimizes spatial filters by maximizing class variances, thereby capturing discriminative neural patterns corresponding to various motor tasks [2]. Based on this, previous work has adopted multi-feature extraction approaches by integrating CSP with other features, such as autoregressive, power spectral density (PSD), and wavelet features, to improve classification performance [3]. SVMs are popular for their suitability for high-dimensional spaces and their ability to generalize well [4]. Combining different learners, for instance, AdaBoost and SVMs, by constructing a learning approach has been proposed to increase classification accuracy [3]. Probabilistic CSP is another extension of CSP that addresses the issue of overfitting [5].

DL methods aim to learn features in a fully automated manner. These methods, however, need extensive amounts of data that are not always available in EEG-based applications. In recent years, efforts have been made to map EEG time series into images known as Gramian Angular Fields (GAF) and to apply DL models, such as generative models, to improve classification accuracy [6]. In light of these facts, this research explores different features and Machine Learning (ML) approaches for the classification of motor imagery EEG.

II. Dataset & Preprocessing

This study utilizes the dataset from BCI Competition 2008 – Graz data set B [7]. The dataset contains EEG recordings from 9 subjects who participated in a cue-based brain-computer interface (BCI) experimental design. The experimental setup required subjects to perform two motor imagery tasks: imagining the movement of the left and right hands. Each subject had 3 recorded sessions. Each session consisted of 120-160

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trials, totaling approximately 400 trials per subject and 3680 trials in total.

During the experiment, subjects were seated facing a computer screen. For a given trial, at $t = 0$ seconds (s), a crosshair appears in the center of the screen along with a warning tone at $t = 2$s. At $t = 3$s, an arrow (pointing left or right) appears on the screen, serving as a visual cue instructing the subject to imagine raising the corresponding hand. This arrow remains displayed for 1.25 seconds. $t = 4$s to $t = 7$s is the 'imagery period' in which neural activity is recorded.

The epoched trials include voltage recordings from 6 channels (3 EEG: C3, Cz, C4 and 3 EOG: ch01, ch02, ch03), with each trial capturing 1000 time points. The complete dataset of 3,680 trials was partitioned using several methods, including a standard 80/20 stratified split, a subject-wise 7:2 split, and a 5-fold cross-validation. Stratified sampling was applied to maintain an equal distribution of left and right-hand imagery across both sets.

III. Methodology

This study evaluates the impact of various feature representations and machine learning models on motor imagery classification performance. The pipeline is divided into traditional machine learning models using kernel Support Vector Machines, Logistic Regression, and Random Forest, and deep learning frameworks (1D and 2D CNN).

A. Feature Engineering and Representation

Flattened signals: Raw EEG trial segments are flattened into 1D vectors, and time-domain statistical metrics are computed to serve as a baseline feature set.

Statistical Features: The mean and variance across the time axis were calculated for each channel and used to evaluate the impact of time-domain data compression. This yielded a reduced set of 12 statistical features per trial, which were standardized prior to modeling.

Bandpower Representations: Frequency domain features are calculated to measure the spectral power in specific neural frequency bands (e.g., mu and beta bands) that are highly correlated with motor imagery tasks.

Gramian Angular Field (GAF) Transformation: The 1D multi-channel EEG time series data are projected into 2D GAF images to benefit from the 2D computer vision architectures. The temporal signal dependencies are encoded in a spatial matrix, thus creating a topological map of the neural activity.

B. Classification Models

Conventional Machine Learning: Logistic Regression, Support Vector Machines, and Random Forest models were trained on the statistical and bandpower features. To account for nonlinear patterns found in the high-dimensional data, the SVM was configured with a Radial Basis Function (RBF) kernel. Model generalization was validated using 5-fold cross-validation.

1D Convolutional Neural Networks (CNN): The 1D CNN architecture is employed to directly model the temporal dynamics of the raw epoched EEG signals, enabling automated feature learning along the time dimension without the need for handcrafted features.

2D Convolutional Neural Networks (CNN): The synthetic 2D GAF images are classified using a 2D CNN framework. EEG classification is treated as an image recognition problem, so that the model can learn complex spatial-temporal representations in the transformed neural data.

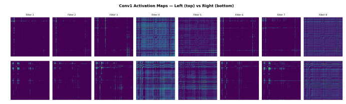

Fig. 1. Visualization of Conv1 activation maps for motor imagery tasks

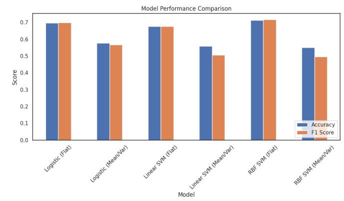

Fig. 2. Performance comparison of traditional machine learning models

IV. Results

Logistic regression achieved the best performance when we use flattened features with an accuracy of 0.697 and an F1-score of 0.699. The performance on means and variance features did not work well; the results indicate that there is important temporal information that was lost.

Support vector machines with RBF kernel showed an improved performance compared to others. The RBF Support Vector Machine while using flattened feature achieved accuracy of 0.713.

Random forest models' performances were improved when a richer feature representation was used. The combination of statistical and band power features achieved accuracy of 0.73 which outperforms other simpler feature sets.

The overall best performance was achieved with the use of the Convolutional Neural Network (CNN). The base line Convolutional Neural Network reached an accuracy of 0.76. The improved Convolutional Neural Network achieved accuracy of 0.765 and an F1 score of 0.767. This shows the advantages of learning features directly from raw signals.

The experimental result above shows how feature representation and model complexity is important in the performance of EEG classification. In this research, we examined machine learning models and deep learning models approaches towards EEG signal classification.

Future improvements would explore Deep Learning models like Recurrent Neural Networks (RNN) and Transformers to automate feature extraction.

V. Conclusion

Feature representation matters more than the model of choice. Preserving temporal structure (flattening) significantly

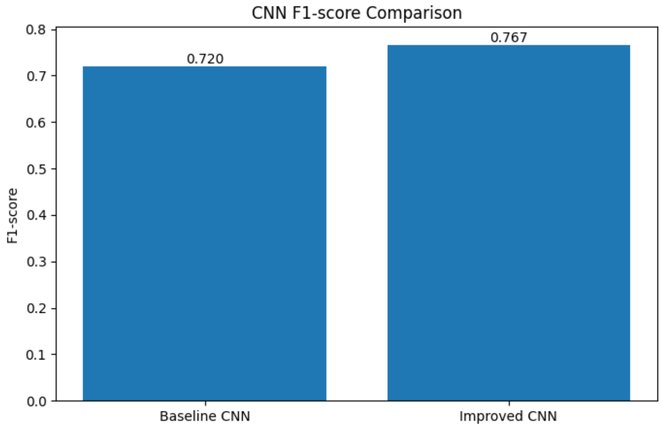

Fig. 3. F1-score comparison between Baseline and Improved 1D CNN models.

outperforms basic statistical aggregation (mean/variance). EEG data contains highly complex, non-linear patterns, which is why 1D CNN in this case outperforms other models.

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